P1954 Reference Design

Open Research Institute

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Who, What, When, Where, Why

Open Source contributions for P1954

- Who?
- Open Research Institute is a nonprofit research and development firm incorporated in California, USA, with international participation.
- All volunteer
- Volunteers, Project Leads, Board of Directors
- Community of ~2000, ~12 projects, 250 active on Slack, ~5-10 developing at any one time.
- https://openresearch.institute

Who, What, When, Where, Why

Open Source contributions for P1954

- What?
- Open Source Digital Radio Research and Development
- All work published at https://github.com/OpenResearchInstitute
- Technical and Regulatory

Who, What, When, Where, Why

Open Source Digital Radio

- When?
- Founded in 2019
- All volunteer, with some active daily and many active weekly or monthly.

Who, What, When, Where, Why

Open Source Digital Radio

- Where?
- Volunteers come from all over the world (most are US-based)
- We demonstrate regularly at conferences and events.
- “It doesn’t work until it works over the air”
- Next demonstration is Friedrichshafen HAMRADIO in June 2026.

Who, What, When, Where, Why

Open Source Digital Radio

- Why?
- We focus on work that 1) is useful and 2) doesn’t *need* to be proprietary
- For workforce development
- For educational development
- For professional development
- For personal development
- To support organizations with Open Source technical or regulatory solutions

Reference Design

Work Products Outline

- Reference design: a working example that bring specifications and requirements to life.
- Our Opulent Voice protocol (MSK) would serve as a reference design command channel (phase 1)
- Our Neptune work (OFDM) would serve as a reference design data channel (phase 2)
- Ardupilot or px4 as flight stack integration (phase 3) hardware interfaces (phase 4) and acceptance tests (phase 4).
- A reference design is not a mandate.
- Goal: demonstrate with Open Source designs that the standard makes basic functional sense. Generate interest, attract useful criticisms, do tests.

Reference Design

What can we bring to the standards effort?

- Reference design: a working example that bring specifications and requirements to life.
- **Our Opulent Voice protocol (MSK) would serve as a reference design command channel.**
- **Our Neptune work (OFDM) would serve as a reference design data channel.**

Reference Design

What can we bring to the standards effort?

- Reference design: a working example that bring specifications and requirements to life.
- **Our Opulent Voice protocol (MSK) would serve as a reference design command channel.**
- https://github.com/OpenResearchInstitute/pluto_msk
- https://github.com/OpenResearchInstitute/dialogus
- https://github.com/OpenResearchInstitute/interlocutor
- https://github.com/OpenResearchInstitute/opv-cxx-demod

Reference Design

What can we bring to the standards effort?

- Reference design: a working example that bring specifications and requirements to life.
- **Our Neptune work (OFDM) would serve as a reference design data channel.**
- https://github.com/OpenResearchInstitute/Neptune
- P1954 provides justification for continuing this work. Win-win.

Reference Design

What can we bring to the standards effort?

- Reference design: a working example that bring specifications and requirements to life.
- **Our Neptune work (OFDM) would serve as a reference design data channel.**
- **ZipCPU/dblclkfft (open source Verilog generator) recommended, multiple other existing open source pieces and parts, no one OFDM design available**